

Realization of Gauge-Reduced 2-Power Residue Matrices

Exact Chebotarev Laws and an Application to Barker Sequences

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Abstract

For a prime $p \equiv 1 \pmod{4}$ such that $\text{ord}_p(2)$ is odd, the 2-primary quotient of $(\mathbb{Z}/p^2\mathbb{Z})^\times$ is cyclic of order $2^{v_2(p-1)}$. Its coordinate is defined only up to multiplication by an odd unit, giving a natural row gauge on matrices of mutual power-residue data. We prove that, after this gauge reduction, quadratic-reciprocity parity and the conditions involving 2 are the only realizability constraints. More precisely, for every fixed realized prefix and every prescribed depth $t \geq 3$, the next-prime transition is an exact prefix-independent Chebotarev law: one fair parity bit per new edge, independent uniform coordinates in the corresponding parity classes, and a common gauge on the new row. The exact-depth slice has density $2/4^t$, and every admissible matrix is realized by infinitely many ordered tuples of distinct primes. The proof computes a Kummer radical with its single exceptional 2-line, determines the maximal abelian subfield by an affine commutator calculation, and identifies the resulting fiber product with quadratic reciprocity. As an application, the reciprocity skeleton used in a companion Barker-sequence census is the exact iterated arithmetic law, not merely a null model. This settles support and iterated-frequency questions while leaving the simultaneous-height regime, and all Barker-specific obstructions, open.

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1 Introduction

Let p be an odd prime and put $t_p = v_2(p-1)$. The reduction map $(\mathbb{Z}/p^2\mathbb{Z})^\times \rightarrow \mathbb{F}_p^\times$ has kernel of odd order, so it induces an isomorphism on maximal 2-power quotients. We write

$$\chi_p : (\mathbb{Z}/p^2\mathbb{Z})^\times \longrightarrow C_{2^{t_p}}$$

for the resulting quotient character, in additive notation. A choice of generator identifies the target with $\mathbb{Z}/2^{t_p}\mathbb{Z}$; changing the generator multiplies the entire row by one odd unit. Thus a matrix $(\chi_{p_i}(p_j))_{i \neq j}$ has an intrinsic row-gauge class even though no individual numerical coordinate is canonical.

We consider the family

$$\mathcal{H} = \{p : p \equiv 1 \pmod{4}, \text{ord}_p(2) \text{ is odd}\}.$$

The terminology “hard prime” is used for this family in the accompanying Barker-sequence code, but is not standard and will not be used in theorem statements. If $p \in \mathcal{H}$, then $\chi_p(2) = 0$ and necessarily $p \equiv 1 \pmod{8}$, so $t_p \geq 3$.

Quadratic reciprocity gives the visible compatibility relation

$$\chi_p(q) \equiv \chi_q(p) \pmod{2} \quad (p, q \in \mathcal{H}, p \neq q). \quad (1)$$

Our main result says that there are no others after row gauge.

Theorem 1.1 (Exact extension law). *Fix distinct primes $P = (p_1, \dots, p_r)$ in $\mathcal{H}$, with $t_i = v_2(p_i - 1)$. Fix $t \geq 3$ and put $N = 2^t$. As q varies over primes in $\mathcal{H}$ of exact depth $v_2(q - 1) = t$, the following conditional distribution exists and is independent of the numerical prefix P :*

- (i) *the parities on the r new edges are independent and fair;*
- (ii) *conditional on those parities, the column coordinates $\chi_{p_i}(q) \in \mathbb{Z}/2^{t_i}\mathbb{Z}$ are independent and uniform in their parity classes;*
- (iii) *the row coordinates $\chi_q(p_i) \in \mathbb{Z}/N\mathbb{Z}$ are independently uniform in the matching parity classes;*
- (iv) *the new row is observed modulo simultaneous multiplication by $(\mathbb{Z}/N\mathbb{Z})^\times$.*

The rational-prime law on gauge orbits is therefore the pushforward of the uniform row-coordinate law. The unconditional density of the slice is

$$\delta_t = \frac{2}{4^t},$$

also independent of P .

Corollary 1.2 (Realization). *For every depth vector $(t_1, \dots, t_k)$ with $t_i \geq 3$, every row-gauge class satisfying (1) is realized by infinitely many ordered tuples of distinct primes in $\mathcal{H}$.*

The theorem is stronger than a support result: it supplies an exact per-prefix transition law. It is also deliberately narrower than a simultaneous-height theorem. All prime bounds may be taken successively, because the conditional law is prefix-independent; nothing here gives a uniform Chebotarev error term as the prefix field grows.

Corollaries 7.2 and 7.3 repackage the theorem as a single object: an exact arithmetic random structure on $\mathcal{H}$. The quadratic-residue parity alone already realizes the countable Rado graph on this thin family; the full structure equips that skeleton with independent geometric depth marks, uniform higher 2-power coordinates, and a row gauge — a gauge-valued 2-adic lift of the prime quadratic-residue Rado graph, together with its exact extension law. The Barker application below is one motif family inside this object.

The classical single-row precedent is Schinzel’s extension of Kummer’s theorem on prescribed power residues [14, 15, 16]. His Theorem 4 is stated for general exponent n , not only odd-prime exponent; over $\mathbb{Q}$ its unrestricted clause includes $n = 2^t$, since the number w of roots of unity in $\mathbb{Q}$ is $2 \not\equiv 0 \pmod{4}$. Applied to multiplicatively independent rational primes, it realizes arbitrary single-row power-residue prescriptions. What it does not supply is the coupled problem here, where the prescribed radicands are themselves the prime moduli, the reciprocal row and column must be realized together, exact depth and the condition on 2 select one Frobenius fiber, and the observable row is defined only up to gauge.

Quadratic residue matrices were characterized by Dummit, Dummit, and Kisilevsky, who also treated cubic and quartic analogues over cyclotomic ground fields [5]. For larger exponents they

identify two complications, not one: higher-power reciprocity itself becomes more complicated, and prime ideals of the cyclotomic base need not be principal. Their closing remark also considers dropping the primary normalization and observes that the resulting larger matrix classes appear less tractable to characterization. The present paper answers that latter difficulty for one specific family: rational-prime radicands at arbitrary 2-power depth. The row-gauge quotient records the discarded generator normalization and restores an exact characterization and transition law. Like their cubic and quartic realization proofs, ours adjoins one prime at a time by Chebotarev; the new content is the quantitative law for that coupled step. We make no claim that this family is equivalent to their general higher-power formulation or that the general $m > 4$ problem is settled. Dummit later developed the d th-power residue-matrix program in the finite-field direction [7]; the present extension instead stays over $\mathbb{Q}$ and moves up the 2-power tower.

DDK also record that direct finite-height counts of their quadratic configuration types can remain far from the exact Chebotarev frequencies, and cite the small-prime bias theorem of Dummit–Granville–Kisilevsky as the explanation [5, 8]. This is the direct precedent for the companion census’s opposite-signed finite-height residuals. It does not resolve them: the DGK theorem treats fixed Dirichlet characters of fixed conductor, whereas the census conditions a nonlinear statistic of mutually varying prime moduli.

The multiquadratic field H also occurs naturally in the theory of minimally tamely ramified extensions. Dummit–Kisilevsky show that the decomposition data of such multiquadratic fields are governed by quadratic residue matrices [6], in the broader minimal-ramification setting of Kisilevsky–Sonn [10]. These results explain the role of H but do not give the higher 2-power intersection $F \cap M = H$ proved below.

The Kummer intersection used below is a specialization of the explicit theory of Perucca–Sgobba–Tronto [13]. We retain a self-contained commutator calculation because it exposes exactly where the exceptional 2-line enters. Their degree uniformity is in the exponents for each fixed finitely generated group G ; the constants and finite case distinction may depend on G . The simultaneous-height problem here requires uniformity on the different, unclaimed axis where $G = \langle 2, p_1, \dots, p_r \rangle$ gains generators and the radicands grow.

2 Gauge-reduced character matrices

Definition 2.1 (Raw and gauge-reduced matrices). Fix depths $\mathbf{t} = (t_1, \dots, t_k)$ and put $N_i = 2^{t_i}$. A raw matrix is an off-diagonal array

$$A = (a_{ij}), \quad a_{ij} \in \mathbb{Z}/N_i\mathbb{Z}.$$

It is *admissible* if $a_{ij} \equiv a_{ji} \pmod{2}$ for every $i \neq j$. The gauge group

$$\mathcal{U}_{\mathbf{t}} = \prod_{i=1}^k (\mathbb{Z}/N_i\mathbb{Z})^\times$$

acts rowwise by $(u_i) \cdot a_{ij} = u_i a_{ij}$. An admissible gauge-reduced matrix is an orbit under this action.

The condition is well-defined on orbits because every gauge multiplier is odd. It is necessary for prime matrices by quadratic reciprocity: the parity of $\chi_p(q)$ records the Legendre symbol (q/p) , and primes in $\mathcal{H}$ are 1 (mod 4). We use the standard power-residue-symbol conventions of Lemmermeyer [11, Chapter 4]; the additive coordinate above records the same cyclic quotient while keeping its generator ambiguity explicit.

Lemma 2.2 (The quotient factors through p). *The reduction map $(\mathbb{Z}/p^2\mathbb{Z})^\times \rightarrow \mathbb{F}_p^\times$ induces an isomorphism on maximal 2-power quotients. Hence $\chi_p(q)$ may be read from the Artin symbol of q in the unique degree- 2^{t_p} subfield of $\mathbb{Q}(\zeta_p)$.*

Proof. The kernel has order p , which is odd. It therefore lies in the unique odd-order subgroup and disappears in the 2-primary quotient. $\square$

3 The governing compositum

Fix a realized prefix $P = (p_1, \dots, p_r)$ and a new depth $t \geq 3$. Put $N = 2^t$ and

$$B = \mathbb{Q}(\zeta_{2N}), \quad B_0 = \mathbb{Q}(\zeta_N), \quad \omega = \zeta_N.$$

For each i , let $F_i \subset \mathbb{Q}(\zeta_{p_i})$ be the unique cyclic subfield of degree 2^{t_i} , and set

$$F = F_1 \cdots F_r, \quad H = \mathbb{Q}(\sqrt{p_1}, \dots, \sqrt{p_r}).$$

Finally choose $\theta_i = p_i^{1/N}$ for $i \geq 1$ and the positive $\theta_0 = 2^{1/N}$, so that $\theta_0^{N/2} = \sqrt{2}$, and set

$$M = B(\theta_0, \theta_1, \dots, \theta_r).$$

The field F carries the old-row/new-column coordinates; M carries the new row and the condition involving 2.

Proposition 3.1 (The radical and its 2-line). *In $B^\times/B^{\times N}$ one has*

$$\langle 2, p_1, \dots, p_r \rangle B^{\times N}/B^{\times N} \cong \mathbb{Z}/(N/2)\mathbb{Z} \oplus (\mathbb{Z}/N\mathbb{Z})^r.$$

Consequently $[M : B] = N^r(N/2)$.

Proof. Suppose $2^{e_0} \prod_i p_i^{e_i} = \beta^N$ in B . The field B is ramified only at 2. At a prime of B over p_j , the normalized valuation of p_j is 1, hence $N \mid e_j$ for every j . Thus no mixed relation survives and only the class of 2 remains.

That class has order at most $N/2$, since

$$2^{N/2} = (\sqrt{2})^N, \quad \sqrt{2} = \zeta_8 + \zeta_8^{-1} \in B.$$

Its order is not smaller. Otherwise $2^{N/4} = \beta^N$ for some β , so $2^{1/4} = \beta\zeta$ for an N th root of unity $\zeta \in B$. This would embed the non-Galois quartic field $\mathbb{Q}(2^{1/4})$ in the abelian extension $B/\mathbb{Q}$, impossible because every subfield of an abelian Galois extension is Galois. Thus the order is exactly $N/2$. $\square$

The exponent is divisible by 8, so the exceptional 2-line behind the Grunwald–Wang special case is a mandatory audit rather than a remote analogy [19]. The Kummer calculation here is over $B \supset \mu_N$, so no local–global existence theorem is being invoked. Instead the possible collapse is computed directly and exhausted: the displayed relation gives the single order drop on the 2-line, while the valuation argument and the lower bound from $\mathbb{Q}(2^{1/4}) \not\subset B$ exclude any second radical relation. This drop will appear below as an even translation coordinate.

4 The maximal abelian subfield

Write $T = \text{Gal}(M/B)$. By Proposition 3.1, an element of T has coordinates $c = (c_0, c_1, \dots, c_r)$ with

$$\tau_c(\theta_i) = \omega^{c_i} \theta_i, \quad c_0 \equiv 0 \pmod{2}.$$

Let $G = \text{Gal}(M/\mathbb{Q})$. For $a \in (\mathbb{Z}/2N\mathbb{Z})^\times$, choose a lift $\tilde{\sigma}_a$ of $\sigma_a \in \text{Gal}(B/\mathbb{Q})$ that fixes the θ_i for $i \geq 1$ and satisfies

$$\tilde{\sigma}_a(\theta_0) = \omega^{u(a)} \theta_0, \quad (-1)^{u(a)} = \epsilon(a) := \frac{\sigma_a(\sqrt{2})}{\sqrt{2}}.$$

Such lifts exist by Proposition 3.1. Moreover

$$\epsilon(a) = 1 \iff a \equiv \pm 1 \pmod{8}. \quad (2)$$

Proposition 4.1 (Maximal abelian subfield). *One has*

$$[G, G] = \text{Gal}(M/BH), \quad M^{\text{ab}} = BH.$$

Proof. Conjugation gives $\tilde{\sigma}_a \tau_c \tilde{\sigma}_a^{-1} = \tau_{ac}$, so commutators with T generate

$$T^2 = \{c_0 \equiv 0 \pmod{4}, c_i \equiv 0 \pmod{2} (i \geq 1)\}.$$

The only missing translation is the second bit on the 2-line. Use left-translation coordinates, with multiplication

$$(a, u)(b, v) = (ab, u + av).$$

$$[(a, u), (b, v)] = (1, (a-1)v - (b-1)u). \quad (3)$$

For $a = 3$ and $b = 5$, both $u(3)$ and $u(5)$ are odd by (2); hence the translation in (3) is 2 (mod 4). Together with T^2 this generates

$$\{c_0 \equiv 0 \pmod{2}, c_i \equiv 0 \pmod{2} (i \geq 1)\} = \text{Gal}(M/BH).$$

The reverse containment follows because $BH/\mathbb{Q}$ is abelian. $\square$

There is an independent external check. Apply Perucca–Sgobba–Tronto [13, Theorem 5.1] to the positive, torsion-free group $\langle 2, p_1, \dots, p_r \rangle$, with their parameters $m = t + 1$ and $n = t$. Their formula gives

$$M \cap \mathbb{Q}^{\text{cyc}} = B(\sqrt{2}, \sqrt{p_1}, \dots, \sqrt{p_r}) = BH.$$

Kronecker–Weber then gives $M^{\text{ab}} = BH$, agreeing with Proposition 4.1.

Corollary 4.2 (The only intersection is reciprocity).

$$F \cap M = H.$$

Proof. Since $F/\mathbb{Q}$ is abelian, $F \cap M \subseteq M^{\text{ab}} = BH$. The field B is totally ramified at 2, whereas F and H are unramified at 2; consequently the inertia-fixed field of BH at 2 is H and $F \cap BH \subseteq H$. Conversely, $\sqrt{p_i}$ lies in both F_i and M , so $H \subseteq F \cap M$. $\square$

The F_i are linearly disjoint over $\mathbb{Q}$: each is ramified only at its own prime p_i , so an intersection with the compositum of the others would be unramified at every finite prime and is therefore $\mathbb{Q}$. Hence

$$\text{Gal}(F/\mathbb{Q}) \cong \prod_{i=1}^r \mathbb{Z}/2^{t_i} \mathbb{Z}. \quad (4)$$

5 Frobenius coordinates and the row gauge

By Corollary 4.2, the usual fiber-product description is

$$\mathrm{Gal}(FM/\mathbb{Q}) \cong \{(\phi, \psi) \in \mathrm{Gal}(F/\mathbb{Q}) \times \mathrm{Gal}(M/\mathbb{Q}) : \phi|_H = \psi|_H\}. \quad (5)$$

Use Lemma 2.2 to choose the i th coordinate in (4) so that the Artin symbol of q is $c_i = \chi_{p_i}(q)$. The restriction of this coordinate to $\mathbb{Q}(\sqrt{p_i})$ is $(-1)^{c_i}$.

The reverse coordinates must be read over B_0 , not over B . If $v_2(q-1) = t$, then

$$q \equiv 1 + N \pmod{2N}.$$

Thus q splits completely in B_0 but has residue degree 2 in B . The extension M/B_0 is Galois: B/B_0 is Galois and B_0 contains μ_N , so every Kummer line used to define M is stable. Choose $\mathfrak{q}_0 \mid q$ in B_0 and a prime above it in M . Since the relative Frobenius fixes ω , write

$$\mathrm{Frob}_{\mathfrak{q}_0}(\theta_i) = \omega^{d_i} \theta_i.$$

Reduction modulo the chosen prime gives

$$\omega^{d_i} \equiv p_i^{(q-1)/N} \pmod{q}.$$

If g is a primitive root modulo q , then $p_i^{(q-1)/N} = (g^{(q-1)/N})^{\chi_q(p_i)}$. The reduction of ω is another generator of the same order- N group. Therefore

$$(d_1, \dots, d_r) = u (\chi_q(p_1), \dots, \chi_q(p_r)) \quad (6)$$

for one odd unit u common to the row.

Changing the prime of B_0 above the rational prime q changes this unit. Changing only the prime of M above a fixed $\mathfrak{q}_0$ does not: an element above σ_{1+N} centralizes T because $1 + N \equiv 1 \pmod{N}$. Thus the ambiguity in (6) is exactly the row gauge, not an additional quotient.

Finally, the condition in (5) is precisely

$$c_i \equiv d_i \pmod{2} \quad (1 \leq i \leq r), \quad (7)$$

which is quadratic reciprocity and nothing more.

6 Proof of the exact extension law

Let $A = \prod_i 2^{t_i} = [F : \mathbb{Q}]$. From the preceding degree calculations,

$$[FM : \mathbb{Q}] = \frac{[F : \mathbb{Q}][M : \mathbb{Q}]}{[H : \mathbb{Q}]} = \frac{A N^{r+2}}{2^{r+1}}. \quad (8)$$

Consider the subset $X_{P,t} \subseteq \mathrm{Gal}(FM/\mathbb{Q})$ defined by

$$\psi|_B = \sigma_{1+N}, \quad \psi(\theta_0) = \theta_0.$$

This is the exact-depth, odd-2-order Frobenius slice.

Lemma 6.1 (Every admissible fiber is nonempty). *Given arbitrary $c_i \in \mathbb{Z}/2^{t_i}\mathbb{Z}$ and $d_i \in \mathbb{Z}/N\mathbb{Z}$ satisfying (7), there is a unique element of $X_{P,t}$ having those coordinates.*

Proof. Choose $\phi \in \text{Gal}(F/\mathbb{Q})$ with coordinates c_i . Define ψ by

$$\psi|_B = \sigma_{1+N}, \quad \psi(\theta_0) = \theta_0, \quad \psi(\theta_i) = \omega^{d_i} \theta_i.$$

Proposition 3.1 says that the displayed radical relations are complete. The only nontrivial compatibility is $\theta_0^{N/2} = \sqrt{2}$, and it holds because $t \geq 3$ gives $1 + N \equiv 1 \pmod{8}$, so $\sigma_{1+N}(\sqrt{2}) = \sqrt{2}$. Thus ψ is an automorphism. On $\sqrt{p_i}$, it acts by $(-1)^{d_i}$; hence (7) is exactly the condition that (ϕ, ψ) belongs to the fiber product (5). Uniqueness is immediate from the generators. $\square$

There are A choices of $c = (c_i)$ and, for each one, $(N/2)^r$ choices of $d = (d_i)$ with matching parity. Therefore

$$|X_{P,t}| = A(N/2)^r. \quad (9)$$

The slice is stable under conjugacy. Indeed, the F component and the restriction to B are abelian, and conjugating an element above σ_{1+N} rescales d by one common unit while keeping the θ_0 coordinate zero. In affine notation the possible translation term is multiplied by $1 - (1 + N)$, which vanishes modulo N .

Chebotarev applied to the union $X_{P,t}$ now gives [12]

$$\delta_t = \frac{|X_{P,t}|}{[FM : \mathbb{Q}]} = \frac{2}{N^2} = \frac{2}{4^t}.$$

More generally, each raw coordinate vector occurs once in $X_{P,t}$, while conjugacy identifies precisely the common row-gauge orbits. Conditional on the slice, Chebotarev therefore gives the pushforward of the uniform raw-coordinate distribution. This proves Theorem 1.1.

For $t = 2$, the two requirements are incompatible: $1 + N \equiv 5 \pmod{8}$ sends $\sqrt{2}$ to $-\sqrt{2}$ and cannot fix θ_0 . This proves the depth floor rather than assuming it. Summing the exact slices gives

$$\sum_{t \geq 3} \frac{2}{4^t} = \frac{1}{24}. \quad (10)$$

The interchange of density and sum is harmless: the upper density of primes with $v_2(p-1) \geq T$ is at most the density 2^{1-T} of $p \equiv 1 \pmod{2^T}$, which tends to zero. Equation (10) agrees with Hasse's density theorem [9]: the odd-order family has density $7/24$, and its $p \equiv 7 \pmod{8}$ part has density $1/4$, leaving $1/24$ in the $p \equiv 1 \pmod{4}$ sector.

Proof of Corollary 1.2. Start with the empty prefix and adjoin the vertices in any chosen order. For a target gauge class, choose a raw representative. If the realized prefix differs by old-row gauges, rescale the corresponding new column coordinates; the transition law is equivariant because odd multiplication preserves each parity class. Lemma 6.1 gives a nonempty conjugacy class for every admissible extension, and Chebotarev supplies infinitely many new primes in it. Such primes are unramified in the compositum and hence distinct from $2, p_1, \dots, p_r$. Induction completes the realization. $\square$

7 The exact iterated matrix law

The extension theorem canonically determines a projectively consistent, exchangeable family of finite probability spaces (Corollary 7.2 below). For a fixed depth vector the construction is as follows. For $i < j$, put

$$E_{ij} = \{(a, b) \in \mathbb{Z}/2^{t_i}\mathbb{Z} \times \mathbb{Z}/2^{t_j}\mathbb{Z} : a \equiv b \pmod{2}\},$$

and define the raw product measure

$$\nu_{\mathbf{t}}^{\text{raw}} = \bigotimes_{i < j} \text{Unif}(E_{ij}). \quad (11)$$

Let $\nu_{\mathbf{t}}$ be its pushforward to row-gauge classes.

Theorem 7.1 (Iterated law). *Fix an ordered depth vector $\mathbf{t}$. Draw primes of these depths from $\mathcal{H}$ and take their bounds to infinity successively, with the last adjoined prime innermost. The resulting distribution of gauge-reduced character matrices is exactly $\nu_{\mathbf{t}}$. It is independent of the adjoining order.*

Proof. For a fixed numerical prefix, Theorem 1.1 supplies the next factor

$$\prod_{i < j} \text{Unif}(E_{ij}),$$

independently of that prefix. Induction therefore gives (11). The kernel is equivariant under every old-row gauge, which rescales the corresponding new column, and under the new-row gauge, which rescales all new-row entries together. Hence choosing a raw representative at an intermediate stage does not affect the final pushforward. Permuting the adjoining order merely permutes the factors in (11). $\square$

This theorem concerns nested limits. It does not assert convergence when all primes are bounded by one common height: the field FM and its discriminant grow with the prefix, and the ordinary Chebotarev theorem used above supplies no uniform error term for that regime.

Two corollaries repackage the depth densities and the iterated law as one unconditional object.

Corollary 7.2 (Unconditional marked product law). *Within $\mathcal{H}$ the exact-depth densities of Theorem 1.1 normalize by (10) to*

$$\Pr(T = t) = \frac{2/4^t}{1/24} = \frac{3}{4^{t-2}}, \quad t \geq 3,$$

a geometric law summing to 1. Draw k primes from $\mathcal{H}$ by successive bounds as in Theorem 7.1, without prescribing depths. Then the depth marks $T_1, \dots, T_k$ are independent with the law above; conditional on $\mathbf{T} = \mathbf{t}$ the raw matrix is distributed as $\nu_{\mathbf{t}}^{\text{raw}}$; explicitly,

$$\Pr(\mathbf{T} = \mathbf{t}, A = a) = \prod_{i=1}^k \frac{3}{4^{t_i-2}} \prod_{1 \leq i < j \leq k} \frac{\mathbf{1}_{\{a_{ij} \equiv a_{ji} \pmod{2}\}}}{2^{t_i+t_j-1}},$$

and the gauge-reduced structure follows the row-gauge pushforward. The family over k is exchangeable and projectively consistent: permuting vertices permutes the factors, and deleting a vertex deletes its depth factor and its incident pair factors. It therefore extends to an exchangeable law on countable gauge-reduced structures. As in Theorem 7.1, every statement is a nested-limit statement; nothing here concerns simultaneous height.

Proof. Depth independence is the prefix-independence of the slice density in Theorem 1.1, normalized by (10); the conditional coordinate law is Theorem 7.1, and the display multiplies the factors $|E_{ij}|^{-1} = 2^{-(t_i+t_j-1)}$. Consistency holds in raw coordinates by the product form, and passes to gauge classes because the odd-unit action on a row is coordinatewise, so restriction to a subset of vertices commutes with the quotient. The Kolmogorov extension theorem then applies. $\square$

Corollary 7.3 (Rado skeleton). *Join $p, q \in \mathcal{H}$ when $\chi_p(q)$ is even — equivalently, by quadratic reciprocity, when p and q are quadratic residues of one another. The resulting countable graph is isomorphic to the Rado graph.*

Proof. Let $U, V \subset \mathcal{H}$ be finite and disjoint. Over the realized prefix $U \cup V$, at any depth $t \geq 3$, Theorem 1.1 makes the new parity bits independent and fair, so the pattern “even toward every vertex of U , odd toward every vertex of V ” is a cylinder of positive density: infinitely many $q \in \mathcal{H}$ are adjacent to all of U and to none of V . A countably infinite graph with this extension property is isomorphic to the Rado graph by back-and-forth [3]. $\square$

The analogous construction on all primes $p \equiv 1 \pmod{4}$ is a classical realization of the Rado graph [3]; the content here is that the extension property survives restriction to the density-1/24 odd-order family, and that this graph is only the parity skeleton of the full gauge-valued structure.

Remark 7.4 (Toward a universal homogeneous gauge structure). Corollary 1.2 proves the one-point extension property in the full gauge-valued signature: every admissible extension of every realized finite substructure is realized by infinitely many primes. Formalizing the class of finite gauge-reduced matrices with its notion of embedding, and identifying the arithmetic structure on $\mathcal{H}$ as the universal homogeneous object of that class — isomorphic to almost every sample of the law of Corollary 7.2 — is left to subsequent work and is not part of the frozen claim set.

8 Application to the Barker reciprocity skeleton

Turyn’s character-sum framework supplies classical necessary conditions for cyclic difference sets and the Barker problem [18]. The companion census [4] studies gauge-invariant predicates of five-vertex character matrices. A pair $\{i, j\}$ is witnessed by $\ell \notin \{i, j\}$ when $a_{\ell i} + a_{\ell j} = 0$. The relevant marked cell imposes global covering and true minimality, no zero off-diagonal entries, an all-odd cofactor row at a marked depth-3 vertex x , and exactly two pairs witnessed by x . Its success event is

$$\sigma_x = \sum_{j \neq x} a_{xj} = 0 \pmod{8}.$$

Both the cell and success are gauge-invariant.

For a full depth vector $\mathbf{t}$, let $A(M, x)$ denote that eligibility event and define the marked-cell intensity ratio

$$R_{\mathbf{t}}^{\text{it}} = \frac{\sum_{x:t_x=3} \nu_{\mathbf{t}}(A(M, x) \wedge \sigma_x = 0)}{\sum_{x:t_x=3} \nu_{\mathbf{t}}(A(M, x))}. \quad (12)$$

The numerator and denominator pool marked cells before division; this is not the expectation of a per-matrix ratio.

An exhaustive exact contraction of (11) in the companion repository gives [4]

$$R_{(3,3,3,3,3)}^{\text{it}} = \frac{1373}{5300} = 0.2590566 \dots, \quad (13)$$

$$R_{(3,3,3,3,4)}^{\text{it}} = \frac{1123}{4215} = 0.2664294 \dots \quad (14)$$

The companion paper uses the four-entry cofactor shorthand $R(3, 3, 3, 3)$ and $R(3, 3, 3, 4)$ for these two quantities. Theorem 7.1 upgrades their interpretation: they are exact iterated arithmetic probabilities, not fitted or Monte Carlo null-model constants.

Corollary 8.1 (Scoped Barker consequence). *No skeleton-admissible pattern is globally absent from tuples of primes in $\mathcal{H}$. Therefore a proposed Barker argument cannot discard such a pattern merely by asserting that its gauge-reduced 2-power residue matrix is arithmetically unrealizable.*

This is a realizability statement only. A realized prime tuple need not be the prime support of a Barker candidate, so a Barker-specific necessary condition may still cut the realized set. Nor does the theorem address nonabelian, class-group, or higher-cohomological data not determined by the matrix. Finally, it does not explain why a skeleton-admissible state may be absent below a finite bound, or why a simultaneous-height census rate may differ from (12). Those are frequency questions outside the support theorem.

9 Computational guardrail and freeze boundary

The proof is algebraic. A deterministic program accompanies it only as a falsification harness. It checks the affine commutator through depths 2–6, the predicted derived subgroup, the exact slice cardinality for multiple prefix ranks, conjugacy as common row scaling, and 56 non-vacuous identifications between the repository’s χ coordinates and the Kummer residue in (6). A sieve over all 1,270,607 primes below $2 \cdot 10^7$ gives the exact-depth counts

t	3	4	5	6	7
#	39711	9939	2536	619	156

and zero at $t = 2$, totaling 53008 primes in the 1 (mod 4) sector. These values reproduce the theorem’s density slices numerically; they are not used as evidence for the proof.

The reproducibility layer is also part of the methodological comparison. The benchmark Barker computations of Mossinghoff and of Borwein–Mossinghoff publish search data and candidate lists and describe their algorithms [1, 2]. They do not describe an executable end-to-end reproduction driver or an independent implementation of the arithmetic primitives. The 2014 sequel itself reports that an earlier implementation of Tarjan’s circuit-enumeration algorithm [17] missed cycles, and that earlier software for Turyn’s test could not resolve some cases [2]. The fail-closed driver, pinned artifacts, regression suite, and clean-room arithmetic path accompanying the census are designed around that documented failure mode. This is a reproducibility advantage, not a claim of priority over the earlier computations.

The program is `barker_k6_bundle/research/realization_checks.py`; its stable output is `_realization_checks.json`. The publication freeze excludes three attractive but logically separate projects:

1. a simultaneous-height convergence or bias theorem;
2. larger Barker censuses or targeted motif enumeration;
3. combinatorial mechanisms for eligibility-weighting effects in the exact skeleton.

Each requires new analytic theory or a new scalable algorithm. None can alter Theorems 1.1 and 7.1, and none is needed for Corollary 8.1 at its stated scope.

Acknowledgment of scope

The central intersection formula is independently covered by the explicit Kummer theory of Perucca–Sgobba–Tronto [13]. The contribution here is the gauge-reduced residue-matrix formulation, the

exact Frobenius slice and transition law, the iterated product law, and their scoped application to the Barker reciprocity skeleton. The paper makes no claim to solve the even Barker conjecture.

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